Text 1

Few creations of big technology capture the imagination like giant dams. Perhaps it is humankind's long suffering at the mercy of flood and drought that makes the idea of forcing the waters to do our bidding so fascinating. But to be fascinated is also, sometimes, to be blind. Several giant dam projects threaten to do more harm than good.

The lesson from dams is that big is not always beautiful. It doesn't help that building a big, powerful dam has become a symbol of achievement for nations and people striving to assert themselves. Egypt's leadership in the Arab world was cemented by the Aswan High Dam. Turkey's bid for First World status includes the giant Ataturk Dam.

But big dams tend not to work as intended. The Aswan Dam, for example, stopped the Nile flooding but deprived Egypt of the fertile silt that floods left – all in return for a giant reservoir of disease which is now so full of silt that it barely generates electricity.

And yet, the myth of controlling the waters persists. This week, in the heart of civilized Europe, Slovaks and Hungarians stopped just short of sending in the troops in their contention over a dam on the Danube. The huge complex will probably have all the usual problems of big dams. But Slovakia is bidding for independence from the Czechs, and now needs a dam to prove itself.

Meanwhile, in India, the World Bank has given the go-ahead to the even more wrong-headed Narmada Dam. And the bank has done this even though its advisors say the dam will cause hardship for the powerless and environmental destruction. The benefits are for the powerful, but they are far from guaranteed.

Proper, scientific study of the impacts of dams and of the cost and benefits of controlling water can help to resolve these conflicts. Hydroelectric power and flood control and irrigation are possible without building monster dams. But when you are dealing with myths, it is hard to be either proper, or scientific. It is time that the world learned the lessons of Aswan. You don't need a dam to be saved.

Text 2

Well, no gain without pain, they say. But what about pain without gain? Everywhere you go in America, you hear tales of corporate revival. What is harder to establish is whether the productivity revolution that businessmen assume they are presiding over is for real.

The official statistics are mildly discouraging. They show that, if you lump manufacturing and services together, productivity has grown on average by 1.2% since 1987. That is somewhat faster than the average during the previous decade. And since 1991, productivity has increased by about 2% a year, which is more than twice the 1978-87 average. The trouble is that part of the recent acceleration is due to the usual rebound that occurs at this point in a business cycle, and so is not conclusive evidence of a revival in the underlying trend. There is, as Robert Rubin, the treasury secretary, says, a "disjunction" between the mass of business anecdote that points to a leap in productivity and the picture reflected by the statistics.

Some of this can be easily explained. New ways of organizing the workplace – all that re-engineering and downsizing – are only one contribution to the overall productivity of an economy, which is driven by many other factors such as joint investment in equipment and machinery, new technology, and investment in education and training. Moreover, most of the changes that companies make are intended to keep them profitable, and this need not always mean increasing productivity: switching to new markets or improving quality can matter just as much.

Two other explanations are more speculative. First, some of the business restructuring of recent years may have been ineptly done. Second, even if it was well done, it may have spread much less widely than people suppose.

Leonard Schlesinger, a Harvard academic and former chief executive of Au Bon Pain, a rapidly growing chain of bakery cafes, says that much "re-engineering" has been crude. In many cases, he believes, the loss of revenue has been greater than the reductions in cost. His colleague, Michael Beer, says that far too many companies have applied re-engineering in a mechanistic fashion, chopping out costs without giving sufficient thought to long-term profitability. BBDO's Al Rosenshine is blunter. He dismisses a lot of the work of re-engineering consultants as mere rubbish – "the worst sort of ambulance chasing."

Text 3

Science has long had an uneasy relationship with other aspects of culture. Think of Galileo's 17th-century trial for his rebelling belief before the Catholic Church or poet William Blake's harsh remarks against the mechanistic worldview of Isaac Newton. The schism between science and the humanities has, if anything, deepened in this century.

Until recently, the scientific community was so powerful that it could afford to ignore its critics – but no longer. As funding for science has declined, scientists have attacked "anti-science" in several books, notably Higher Superstition, by Paul R. Gross, a biologist at the University of Virginia, and Norman Levitt, a mathematician at Rutgers University; and The Demon-Haunted World, by Carl Sagan of Cornell University.

Defenders of science have also voiced their concerns at meetings such as "The Flight from Science and Reason," held in New York City in 1995, and "Science in the Age of (Mis) information," which assembled last June near Buffalo.

Anti-science clearly means different things to different people. Gross and Levitt find fault primarily with sociologists, philosophers and other academics who have questioned science's objectivity. Sagan is more concerned with those who believe in ghosts, creationism and other phenomena that contradict the scientific worldview.

A survey of news stories in 1996 reveals that the anti-science tag has been attached to many other groups as well, from authorities who advocated the elimination of the last remaining stocks of smallpox virus to Republicans who advocated decreased funding for basic research.

Few would dispute that the term applies to the Unabomber, whose manifesto, published in 1995, scorns science and longs for return to a pre-technological utopia. But surely that does not mean environmentalists concerned about uncontrolled industrial growth are anti-science, as an essay in US News & World Report last May seemed to suggest.

The environmentalists, inevitably, respond to such critics. The true enemies of science, argues Paul Ehrlich of Stanford University, a pioneer of environmental studies, are those who question the evidence supporting global warming, the depletion of the ozone layer and other consequences of industrial growth.

Indeed, some observers fear that the anti-science epithet is in danger of becoming meaningless. "The term 'anti-science' can lump together too many, quite different things," notes Harvard University philosopher Gerald Holton in his 1993 work Science and Anti-Science. "They have in common only one thing that they tend to annoy or threaten those who regard themselves as more enlightened."

Text 4

Emerging from the 1980 census is the picture of a nation developing more and more regional competition, as population growth in the Northeast and Midwest reaches a near standstill.

This development – and its strong implications for US politics and economy in years ahead – has enthroned the South as America's most densely populated region for the first time in the history of the nation's head counting.

Altogether, the US population rose in the 1970s by 23.2 million people – numerically the third-largest growth ever recorded in a single decade. Even so, that gain adds up to only 11.4 percent, lowest in American annual records except for the Depression years.

Americans have been migrating south and west in larger numbers since World War II, and the pattern still prevails.

Three sun-belt states – Florida, Texas and California – together had nearly 10 million more people in 1980 than a decade earlier. Among large cities, San Diego moved from 14th to 8th and San Antonio from 15th to 10th – with Cleveland and Washington. D. C., dropping out of the top 10.

Not all that shift can be attributed to the movement out of the snow belt, census officials say. Nonstop waves of immigrants played a role, too – and so did bigger crops of babies as yesterday's "baby boom" generation reached its child bearing years.

Moreover, demographers see the continuing shift south and west as joined by a related but newer phenomenon: More and more, Americans apparently are looking not just for places with more jobs but with fewer people, too. Some instances:

•Regionally, the Rocky Mountain states reported the most rapid growth rate – 37.1 percent since 1970 in a vast area with only 5 percent of the US population.

•Among states, Nevada and Arizona grew fastest of all: 63.5 and 53.1 percent respectively. Except for Florida and Texas, the top 10 in rate of growth is composed of Western states with 7.5 million people – about 9 per square mile.

The flight from overcrowdedness affects the migration from snow belt to more bearable climates. Nowhere do 1980 census statistics dramatize more the American search for spacious living than in the Far West. There, California added 3.7 million to its population in the 1970s, more than any other state.

In that decade, however, large numbers also migrated from California, mostly to other parts of the West. Often they chose – and still are choosing – somewhat colder climates such as Oregon, Idaho and Alaska in order to escape smog, crime and other plagues of urbanization in the Golden State.

As a result, California's growth rate dropped during the 1970s, to 18.5 percent – little more than two thirds the 1960s' growth figure and considerably below that of other Western states.